

Operator-Adapted Reproducing Kernel Geometries for Structure-Preserving Representation Learning

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Abstract

The representation geometry determines which transformations appear to preserve, distort, or separate the represented objects. In most representation-learning settings, this geometry is fixed in advance or learned from data, while the action of an operator is subsequently studied within it. We consider the inverse problem by asking, given a linear operator, can one construct a reproducing kernel Hilbert space (RKHS) geometry in which its action is exactly norm preserving? We establish necessary and sufficient conditions for the existence of such an operator-adapted RKHS in finite-dimensional function spaces and characterize the complete family of reproducing kernels realizing this invariance. The resulting classification shows that existence is governed by the spectral structure of the operator, while the spectral multiplicities determine the degrees of freedom available in the invariant representation geometry. In particular, an operator that is not unitary under the original geometry may become unitary after an appropriate change of RKHS geometry, whereas operators with incompatible spectral structure admit no such positive-definite realization. These results provide a mathematical framework for constructing representations from the prescribed operator structure and suggest an alternative principle for structure-preserving representation learning where rather than modifying an operator to respect a predetermined geometry, one may construct the geometry to respect the operator.

1 Introduction

The performance of any Artificial Intelligence and machine learning model is heavily dependent on the choice of data representation [2] that is technically often referred to as *features* [2]. In this concern, representation learning plays a monumental role by providing a priori details from the aspect of several general purposes such as probability distributions embedding [7], spatial and temporal coherence, manifold, semi-supervised learning, sparsity, etc.

Kernel machine learning [5, 18] offers a promising path for representation learning of features for high dimensional data. Essentially, the high dimensional data gets encoded in the inner-product between the infinite dimensional nonlinear kernel feature space over the underlying Hilbert space. These specialized function spaces corresponding to kernels are called *reproducing kernel Hilbert spaces* [1] (RKHS). Besides being able to handle the huge dimension of data, RKHS also offers a unique notion of representation geometry that also encapsulates data similarity, distance and structural variation by getting embedded in it through the kernel representation.

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Linear transformations such as the Koopman (*composition*) operators [12], are unitary in L^2 [17] under measure-preserving invertible dynamics [13], making them desirable operator learning for achieving robust prediction, control, regression and etc. in various scientific domains [21, 4, 9]. In general, it is immediate to observe that the same *desirable* notion cannot be offered within the RKHS.

In this paper, we are *motivated* by this very aim of adapting RKHS for which the operator norm geometry is already known. In particular, we establish an existence and classification theory for such operator-adapted RKHS geometries. Exact preservation is possible precisely when the matrix representation of the operator is diagonalizable with spectrum on the unit circle. When this condition is satisfied, the complete family of admissible geometries is determined by the spectral decomposition of the operator. We further show how the same principle produces an operator-adapted Gaussian RKHS by replacing the Euclidean quadratic form in the Gaussian radial basis function kernel [5, 20] with an invariant positive-definite metric. This construction permits transformations that are not unitary in Euclidean geometry to induce unitary Koopman composition operators in the adapted RKHS whenever the corresponding invariant metric exists.

Problem statement. Let X be a nonempty subset inside $\mathbb{R}^d$, and let $\mathbb{C}^X := \{f \mid f : X \rightarrow \mathbb{C}\}$. Let $1 \leq j \leq m$ be natural indexes and $\psi_j : X \rightarrow \mathbb{C}$ be a basis function with $\mathcal{V} := \text{span}\{\psi_1, \dots, \psi_m\} \subseteq \mathbb{C}^X$. Let $T : \mathcal{V} \rightarrow \mathcal{V}$ be a linear operator. We are interested in inner products on $\mathcal{V}$ for which $(\mathcal{V}, \langle \cdot, \cdot \rangle_{\mathcal{H}_k})$ is an RKHS $\mathcal{H}_k$ with a reproducing kernel function k such that T acts isometrically:

$$\|Tf\|_{\mathcal{H}_k} = \|f\|_{\mathcal{H}_k} \quad \forall f \in \mathcal{V}.$$

Accordingly, we address three questions:- (i) under what conditions does such an RKHS geometry exist; (ii) when it exists, how can the complete family of admissible kernels be characterized; and (iii) how can the resulting invariant geometry be realized within a standard kernel family so that a prescribed transformation induces a unitary composition action?

1.1 Related Work

Transformation-invariant kernels [14, 18] have been studied for geometric actions such as translations, rotations, and dilations, while group-averaging constructions have been used to obtain invariant feature maps and associated invariant RKHSs [19, 16]. In these approaches, the transformation class is typically specified first, and invariance is built into the kernel or feature representation.

Recent results characterize when mappings of the input space induce bounded composition operators in RKHSs associated with analytic positive definite functions; in broad classes, boundedness can itself force the underlying map to be affine [10, 8]. Such results typically begin with a fixed RKHS and ask which composition operators [6] it admits. Our finite-dimensional problem reverses this direction, that is, the operator is fixed, while the Hilbertian geometry is unknown and is to be constructed so that the operator becomes isometric.

Kernel methods have also become important in Koopman operator theory, where the RKHS structure is used to approximate Koopman operators, eigenfunctions, or observables and to derive approximation guarantees [21, 11]. These works generally assume or select an RKHS and then study the Koopman operator within that space. Our construction instead asks whether the geometry itself can be chosen from the operator.

2 Operator-Adapted Reproducing Kernel Geometries

2.1 RKHS

Let $k : X \times X \rightarrow \mathbb{C}$ be a continuous Hermitian positive definite kernel denoted by $\succ 0$. At $x \in X$, define the canonical feature $\phi_z(x) = k(x, z)$ centered at $z \in X$. Then, the linear space given by $\text{span}\{\phi_z : z \in X\}$ extends by completion to a Hilbert space $\mathcal{H}_k$, called the *reproducing kernel Hilbert space* (RKHS) [5] generated by $k(\cdot, \cdot)$. For all functions $f \in \mathcal{H}_k$, the following properties hold true at $x \in X$:

$$\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} = f(x) \tag{1a}$$

$$\langle k(\cdot, y), k(\cdot, x) \rangle_{\mathcal{H}_k} = k(x, y) \tag{1b}$$

$$|f(x)| \leq \|f\|_{\mathcal{H}_k} \cdot (k(x, x))^{1/2}. \tag{1c}$$

The property shown in (1a) is called the *reproducing property* of $k(\cdot, x)$ and therefore $k(\cdot, x)$ is called the reproducing kernel of $\mathcal{H}_k$. When $f = k(\cdot, y)$ in (1a), we get (1b) and finally (1c) shows the boundedness of the *evaluation functional* [15, 1] $\mathcal{E}_x(f) := f(x)$ for $f \in \mathcal{H}_k$ in $x \in X$.

2.2 Operator Adapted RKHS

Recall that X is a nonempty set and let $\mathcal{V} = \text{span}\{\psi_1, \dots, \psi_m\}$ be a m -dimensional space of complex-valued functions on X , where $\psi_1, \dots, \psi_m$ are linearly independent such that $\dim \mathcal{V} = m$. Define the vector valued representation as follows through linearly independent basis functions ψ_j :

$$\Psi(x)^* = [\psi_1(x) \quad \dots \quad \psi_m(x)], \quad x \in X,$$

where $\bullet^*$ implies the Hermitian-adjoint. For $c := [c_j]_{j=1}^m \in \mathbb{C}^m$, write $f_c(x) = \Psi(x)^* c = \sum_{j=1}^m c_j \psi_j(x)$. Let $T : \mathcal{V} \rightarrow \mathcal{V}$ be a linear operator. Relative to the prescribed basis, let $M \in \mathbb{C}^{m \times m}$ denote its matrix representation, so that $Tf_c = f_{Mc}$.

We write $\mathbb{H}_{++}^m := \{G \in \mathbb{C}^{m \times m} : G = G^*, G \succ 0\}$ for the open convex cone of Hermitian positive-definite matrices [3]. For $G \in \mathbb{H}_{++}^m$, we defined the induced inner-product and norm as follows:

$$\begin{aligned} \langle f_c, f_d \rangle_G &= d^* G c \\ \|f_c\|_G^2 &= c^* G c. \end{aligned} \tag{2}$$

In this section, we characterize those RKHS geometries on $\mathcal{V}$ for which T acts isometrically. The following proposition provides the RKHS realization whose inner product is given in (2).

Proposition 2.1. *For every $G \in \mathbb{H}_{++}^m$, the Hilbert space $(\mathcal{V}, \langle \cdot, \cdot \rangle_G)$ is an RKHS denoted by $\mathcal{H}_{k_G}$ with the reproducing kernel given as: $k_G(x, y) = \Psi(x)^* G^{-1} \Psi(y)$, $x, y \in X$.*

We also have uniqueness of kernel parametrization, meaning if $k_{G_1}(x, y) = k_{G_2}(x, y)$ for all $x, y \in X$, then $G_1 = G_2$. Refer Proposition B.2 for this result. Now that we have established the preliminary foundation structure of RKHS $\mathcal{H}_{k_G}$ in Proposition 2.1, we then establish the existence of an invariant RKHS geometry for a linear operator T as follows.

Theorem 2.2. *Let $T : \mathcal{V} \rightarrow \mathcal{V}$ and $G \in \mathbb{H}_{++}^m$. Consider RKHS $\mathcal{H}_{k_G}$ as given in Proposition 2.1, then the following are equivalent:*

1. T acts isometrically on $\mathcal{H}_{k_G}$;
2. $\|Tf\|_{\mathcal{H}_{k_G}} = \|f\|_{\mathcal{H}_{k_G}} \quad \forall f \in \mathcal{V}$;
3. $M^* G M = G$.

Consequently, $\exists G \in \mathbb{H}_{++}^m$ for which T acts isometrically if and only if M is similar to a unitary matrix. Equivalently, M is diagonalizable and $\sigma(M) \subset \mathbb{T}$, where $\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}$.

Remark 2.3. *Condition $\sigma(M) \subset \mathbb{T}$ alone is not sufficient. A nontrivial Jordan block associated with a unimodular eigenvalue cannot satisfy $M^* G M = G$ for any $G \succ 0$. Thus, diagonalizability constitutes an additional obstruction to the existence of an exact invariant positive-definite geometry.*

For $M \in \mathbb{C}^{m \times m}$, we let the collection of all admissible matrices $G \in \mathbb{H}_{++}^m$ such that $M^* G M = G$ holds; this is given by $\mathcal{G}(M) := \{G \in \mathbb{H}_{++}^m : M^* G M = G\}$. The following theorem helps to provide the classification of invariant geometries whose equivalent conditions were provided in Theorem 2.2. Define the family of operator-adapted kernels by $\mathcal{R}_\Psi(T) := \{k_G : G \in \mathcal{G}(M)\}$.

Theorem 2.4. *Assume that M is diagonalizable and its spectrum satisfies $\sigma(M) \subset \mathbb{T}$. Let $\mu_1, \dots, \mu_r$ denote the distinct eigenvalues of M , with multiplicities $m_1, \dots, m_r$, respectively. Choose an eigenvector matrix S such that $M = SDS^{-1}$, where $D = \text{diag}(\mu_1 I_{m_1}, \dots, \mu_r I_{m_r})$. Then $G \in \mathbb{H}_{++}^m$ belongs to $\mathcal{G}(M)$ if and only if $G = S^{-*} \text{diag}(H_1, \dots, H_r) S^{-1}$, where $H_j \in \mathbb{H}_{++}^{m_j}$, $j = 1, \dots, r$. Here, $S^{-*} := (S^{-1})^* = (S^*)^{-1}$.*

We take advantage of the spectral structure of $M \in \mathbb{C}^{m \times m}$ to classify exactly which RKHS geometries and, consequently, which RKHS kernels are compatible with operator invariance; this classification is completed in the following corollary.

Corollary 2.5. According to the hypotheses of [Theorem 2.4](#), the complete family of reproducing kernels for which T acts isometrically is $k_{\mathbf{H}}(x, y) = \Psi(x)^* S \operatorname{diag}(H_1^{-1}, \dots, H_r^{-1}) S^* \Psi(y)$, where $(H_1, \dots, H_r) \in \prod_{j=1}^r \mathbb{H}_{++}^{m_j}$.

Corollary 2.6. According to the hypotheses of [Theorem 2.4](#), $\dim_{\mathbb{R}} \mathcal{G}(M) = \dim_{\mathbb{R}} \mathfrak{K}_{\Psi}(T) = \sum_{j=1}^r m_j^2$. Thus, the spectral multiplicity determines the degrees of freedom in the family of operator-adapted RKHS kernels.

2.3 Numerical Verification of Operator-Adapted Geometry

We illustrate the construction on a $d = 20$ linear representation operator M that is similar to a unitary matrix but is not unitary in the native Euclidean geometry. Thus, although its eigenvalues lie on $\mathbb{T}$, its singular values differ substantially from one, showing that spectral preservation alone does not imply preservation of the ambient representation geometry. We construct $G \succ 0$ satisfying $M^* G M = G$. Equivalently, the similarity-transformed operator $\tilde{M} = G^{1/2} M G^{-1/2}$ is unitary.

[Figure 1](#) illustrates the resulting change in geometry. The spectrum is unchanged under the similarity transformation ([Figure 1a](#)), while the singular values of M collapse to one despite the substantial singular value variation of M in the Euclidean metric ([Figure 1b](#)). Then, we compare the Gaussian radial basis function kernel $k_0(x, y) = \exp\left(-\frac{\|x-y\|_2^2}{2\gamma^2}\right)$ with its operator-adapted counterpart $k_G(x, y) = \exp\left(-\frac{(x-y)^* G (x-y)}{2\gamma^2}\right)$ with the kernel bandwidth $\gamma = 3$.

To test invariance directly, we sample representations $z_1, \dots, z_N \in \mathbb{R}^{20}$, with $N = 120$ and compare pairwise kernel values before and after application of M . For a kernel k , we define the entrywise invariance discrepancy as $\Delta_k(i, j) = |k(Mz_i, Mz_j) - k(z_i, z_j)|$. Thus, $\Delta_k(i, j) = 0$ for every pair (i, j) whenever k is exactly invariant under M . We summarize the discrepancy by the normalized error given as follows:

$$\varepsilon_k := \sqrt{\frac{\sum_{i,j=1}^N \Delta_k^2(i, j)}{\sum_{i,j=1}^N |k(z_i, z_j)|^2}} = \sqrt{\frac{\sum_{i,j=1}^N |k(Mz_i, Mz_j) - k(z_i, z_j)|^2}{\sum_{i,j=1}^N |k(z_i, z_j)|^2}}.$$

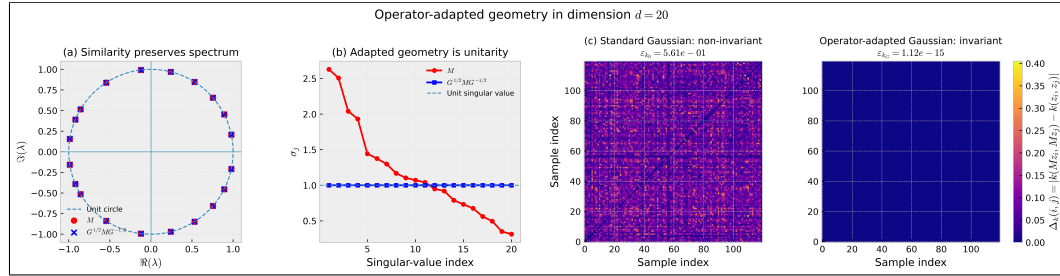


Figure 1: Operator adaptation converts a spectrally stable but geometrically non-isometric representation into an *exact* isometry.

Using a common color scale, [Figure 1c-d](#) demonstrates the collapse of the invariance error from $\varepsilon_{k_0} = 5.61 \times 10^{-1}$ to $\varepsilon_{k_G} = 1.12 \times 10^{-15}$, while the operator spectrum remains unchanged. Thus, adapting the RKHS metric does not alter the operator spectrum but changes the geometry in which the same operator acts from non-isometric to isometric.

3 Conclusion

We formulated operator-adapted RKHS construction as an inverse geometric problem and characterized exactly when a finite-dimensional linear operator admits an invariant positive-definite geometry. Existence is equivalent to similarity to a unitary matrix, while spectral multiplicities parameterize the complete family of invariant RKHS kernels. The same principle yields operator-adapted Gaussian kernels that preserve pairwise similarity even when the underlying transformation is non-unitary.

in Euclidean geometry. Thus, rather than modifying an operator to fit a prescribed representation geometry, the geometry itself can be constructed to respect the operator.

As a limitation of our proposed research work, the present results concern finite-dimensional function spaces. The corresponding problem in infinite-dimensional RKHSs leads to the operator equation $T^*GT = G$, with G a bounded, positive, and boundedly invertible operator, together with the question of how such invariant Hilbert geometries are realized at the level of reproducing kernels. Establishing an intrinsic classification in this setting is a natural direction beyond the finite-dimensional theory developed here.

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A Proofs

A.1 Proof for Proposition 2.1

Proof. Fix $x \in X$ and define $k_{G,x} = f_{G^{-1}\Psi(x)}$. Then, for every $f_c \in \mathcal{V}$,

$$\begin{aligned}\langle f_c, k_{G,x} \rangle_G &= \Psi(x)^* G^{-1} G c \\ &= \Psi(x)^* c \\ &= f_c(x).\end{aligned}$$

Thus $k_{G,x}$ represents point evaluation at x , and hence $(\mathcal{V}, \langle \cdot, \cdot \rangle_G)$ is an RKHS satisfying $k_G(x, y) = \Psi(x)^* G^{-1} \Psi(y)$. $\square$

A.2 Proof for Theorem 2.2

Proof. For $f_c \in \mathcal{V}$, $\|Tf_c\|_G^2 = \|f_{Mc}\|_G^2 = c^* M^* G M c$. Hence

$$\|Tf_c\|_G^2 = \|f_c\|_G^2 \quad \text{for every } c \in \mathbb{C}^m \iff M^* G M = G.$$

Suppose first that $G \succ 0$ satisfies $M^* G M = G$. Define $U = G^{1/2} M G^{-1/2}$. Then

$$U^* U = G^{-1/2} M^* G M G^{-1/2} = I,$$

so U is unitary. Therefore, M is similar to a unitary matrix.

Conversely, suppose $M = S D S^{-1}$, where $D = \text{diag}(\lambda_1, \dots, \lambda_m)$, $|\lambda_j| = 1$. Set $G = S^{-*} S^{-1}$. Then $G \succ 0$ and

$$\begin{aligned}M^* G M &= S^{-*} D^* D S^{-1} \\ &= S^{-*} S^{-1} \\ &= G.\end{aligned}$$

The spectral characterization follows. $\square$

A.3 Proof for Theorem 2.4

Proof. Write $G = S^{-*} H S^{-1}$. Then $M^* G M = G \implies D^* H D = H$. Writing $D = \text{diag}(\lambda_1, \dots, \lambda_m)$, the (i, j) entry of this equation is $\bar{\lambda}_i \lambda_j H_{ij} = H_{ij}$. Thus $(\bar{\lambda}_i \lambda_j - 1) H_{ij} = 0$.

Since $|\lambda_i| = |\lambda_j| = 1$, $\bar{\lambda}_i \lambda_j = 1 \iff \lambda_i = \lambda_j$. Hence $H_{ij} = 0$ whenever $\lambda_i \neq \lambda_j$. Therefore H is block diagonal with respect to the distinct eigenspaces $H = \text{diag}(H_1, \dots, H_r)$, where each H_j is Hermitian positive definite. The converse follows immediately by substitution. $\square$

A.4 Proof for Corollary 2.5

Proof. By Proposition 2.1, $k_G(x, y) = \Psi(x)^* G^{-1} \Psi(y)$. For $G = S^{-*} \text{diag}(H_1, \dots, H_r) S^{-1}$, we have

$$G^{-1} = S \text{diag}(H_1^{-1}, \dots, H_r^{-1}) S^*.$$

The result follows from Theorem 2.4. $\square$

A.5 Proof for Corollary 2.6

Proof. Theorem 2.4 identifies $\mathcal{G}(M)$ with the product $\prod_{j=1}^r \mathbb{H}_{++}^{m_j}$. The real vector space of $m_j \times m_j$ Hermitian matrices has dimension m_j^2 , and the positive-definite cone is an open subset thereof. Hence $\dim_{\mathbb{R}} \mathcal{G}(M) = \sum_{j=1}^r m_j^2$. In addition to this, the second equality follows from the injective metric-kernel correspondence. $\square$

B Other important results

Corollary B.1 (Simple spectrum). *Suppose that M has m distinct eigenvalues and $\sigma(M) \subset \mathbb{T}$. Then every invariant geometry has the form*

$$G_{\mathbf{h}} = S^{-*} \text{diag}(h_1, \dots, h_m) S^{-1}, \quad h_j > 0,$$

and the corresponding reproducing kernels are

$$k_{\mathbf{h}}(x, y) = \Psi(x)^* S \text{diag}(h_1^{-1}, \dots, h_m^{-1}) S^* \Psi(y).$$

B.1 Uniqueness of the kernel parameterization

Proposition B.2 (Injectivity of the metric–kernel correspondence). *Suppose that $\psi_1, \dots, \psi_m$ are linearly independent. If $k_{G_1}(x, y) = k_{G_2}(x, y)$ for all $x, y \in X$, then $G_1 = G_2$.*

Proof. The equality of the kernels gives $\Psi(x)^* (G_1^{-1} - G_2^{-1}) \Psi(y) = 0$ for all $x, y \in X$. Since $\psi_1, \dots, \psi_m$ are linearly independent, $\text{span}\{\Psi(x) : x \in X\} = \mathbb{C}^m$. It follows that $G_1^{-1} - G_2^{-1} = 0 \implies G_1 = G_2$. $\square$

In particular, if the spectrum is simple, then $\dim_{\mathbb{R}} \mathcal{G}(M) = m$. At the opposite extreme, if $M = \lambda I$, $|\lambda| = 1$, then $\mathcal{G}(M) = \mathbb{H}_{++}^m$, $\dim_{\mathbb{R}} \mathcal{G}(M) = m^2$. Eventually, the combined application of [Proposition B.2](#) and [Theorem 2.4](#) produces the bijective parameterization

$$\mathfrak{K}_{\Psi}(T) \cong \prod_{j=1}^r \mathbb{H}_{++}^{m_j}.$$

B.2 An operator-adapted Gaussian RKHS

The preceding finite-dimensional characterization also induces operator-adapted kernels within standard kernel families. We illustrate this observation for the Gaussian radial basis function kernel.

Let $Q \in \mathbb{C}^{d \times d}$ be unitary and let $S \in \mathbb{C}^{d \times d}$ be invertible. Define $A = SQS^{-1}$. Although A is similar to a unitary matrix, it does not need to be unitary with respect to the Euclidean inner product. In particular, for a non-unitary choice of S , one generally has

$$A^* A \neq I.$$

Define $G = S^{-*} S^{-1}$. Then $G \succ 0$, and

$$\begin{aligned} A^* G A &= (SQS^{-1})^* S^{-*} S^{-1} (SQS^{-1}) \\ &= S^{-*} Q^* S^* S^{-*} S^{-1} S Q S^{-1} \\ &= S^{-*} Q^* Q S^{-1} \\ &= S^{-*} S^{-1} \\ &= G. \end{aligned}$$

Thus A is unitary with respect to the inner product induced by G .

Consider the Gaussian kernel

$$k_G(x, y) = \exp \left(-\frac{(x - y)^* G (x - y)}{2\sigma^2} \right), \quad x, y \in \mathbb{R}^d,$$

where $\sigma > 0$. Since $A^* G A = G$,

$$\begin{aligned} k_G(Ax, Ay) &= \exp \left(-\frac{(A(x - y))^* G A(x - y)}{2\sigma^2} \right) \\ &= \exp \left(-\frac{(x - y)^* A^* G A(x - y)}{2\sigma^2} \right) \\ &= k_G(x, y). \end{aligned}$$

Hence the kernel is invariant under the transformation A .

Let $\mathcal{K}_A$ denote the Koopman operator

$$(\mathcal{K}_A f)(x) = f(Ax).$$

For the kernel sections $k_x = k_G(\cdot, x)$, the adjoint satisfies

$$\mathcal{K}_A^* k_x = k_{Ax}.$$

Therefore

$$\begin{aligned} \langle \mathcal{K}_A^* k_x, \mathcal{K}_A^* k_y \rangle_{\mathcal{H}_{k_G}} &= \langle k_{Ax}, k_{Ay} \rangle_{\mathcal{H}_{k_G}} \\ &= k_G(Ay, Ax) \\ &= k_G(y, x) \\ &= \langle k_x, k_y \rangle_{\mathcal{H}_{k_G}}. \end{aligned}$$

Since the linear span of the kernel sections is dense in $\mathcal{H}_{k_G}$, $\mathcal{K}_A^*$ extends isometrically to the whole space. As A is invertible, the induced composition action is surjective, and the corresponding Koopman operator is unitary on $\mathcal{H}_{k_G}$.

By contrast, the isotropic Gaussian radial basis function kernel [5] $k_I(x, y) = \exp\left(-\frac{\|x - y\|_2^2}{2\sigma^2}\right)$ satisfies

$$k_I(Ax, Ay) = k_I(x, y)$$

for all x, y only when $A^*A = I$. Thus a transformation that is not Euclidean-unitary need not preserve the standard Gaussian RKHS, while the operator-adapted metric G produces a Gaussian RKHS in which the same transformation acts unitarily.

B.3 Normalization

The invariance relation is homogeneous. Indeed, if $G \in \mathcal{G}(M)$, then $cG \in \mathcal{G}(M)$ for every $c > 0$. Moreover,

$$k_{cG}(x, y) = \frac{1}{c} k_G(x, y).$$

Thus positive scalar multiplication changes the normalization of the Hilbert norm while preserving the underlying function space. It is therefore natural to consider invariant geometries modulo the equivalence relation

$$G_1 \sim G_2 \iff G_2 = cG_1 \text{ for some } c > 0.$$

Define

$$\mathfrak{M}_\Psi(T) = \mathcal{G}(M)/\mathbb{R}_{>0}.$$

Equivalently, one may choose a representative of each equivalence class by imposing, for example, $\text{tr}(G) = 1$. The resulting normalized family has real dimension

$$\dim_{\mathbb{R}} \mathfrak{M}_\Psi(T) = \sum_{j=1}^r m_j^2 - 1.$$

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